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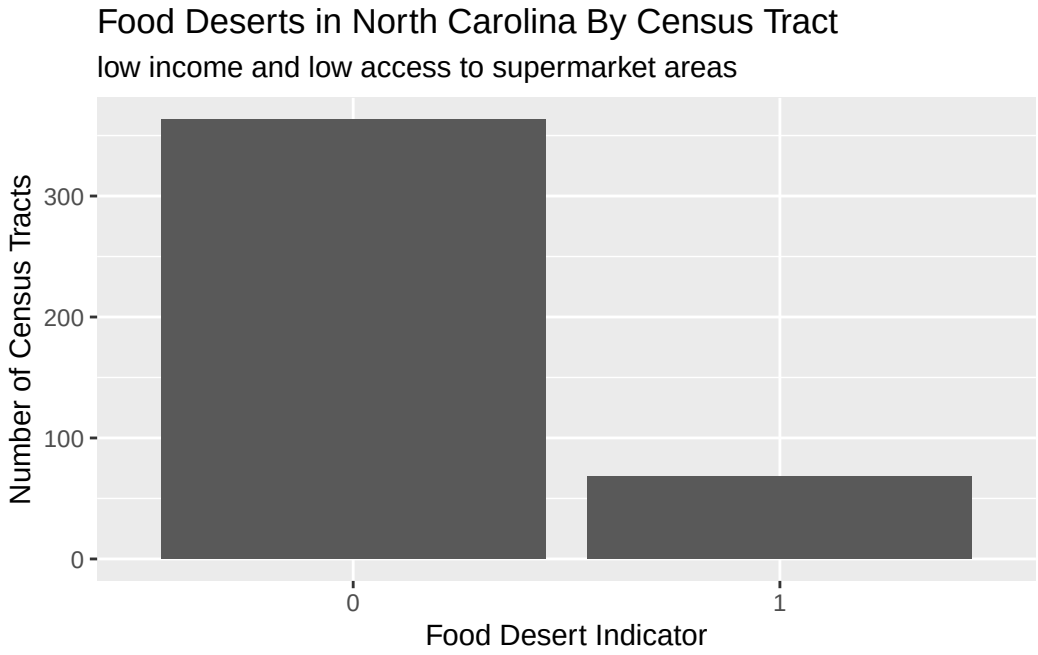
> **!** Important
>
> Before you submit, make sure your code chunks are turned off with `echo: false` and there are no warnings or messages with `warning: false` and `message: false` in the YAML.

Exploratory Data Analysis

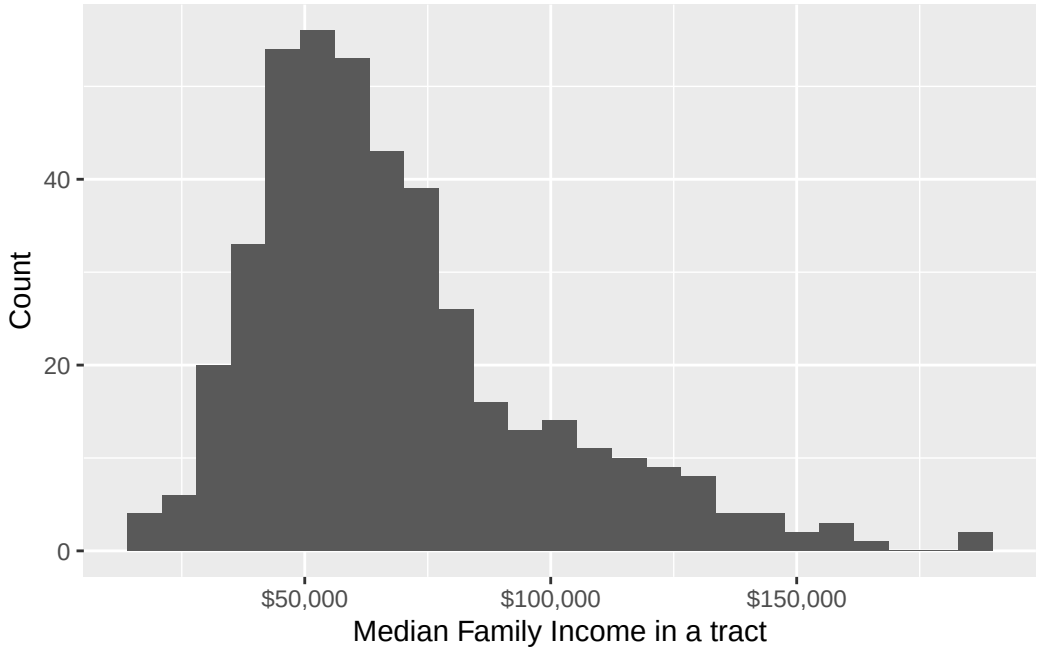
in what ways do the poverty rate of a census tract, the number of households receiving SNAP benefits, the number of households without a vehicle, and urban/rural classification, median income influence how many low-income people in a census tract live in a low food access area?

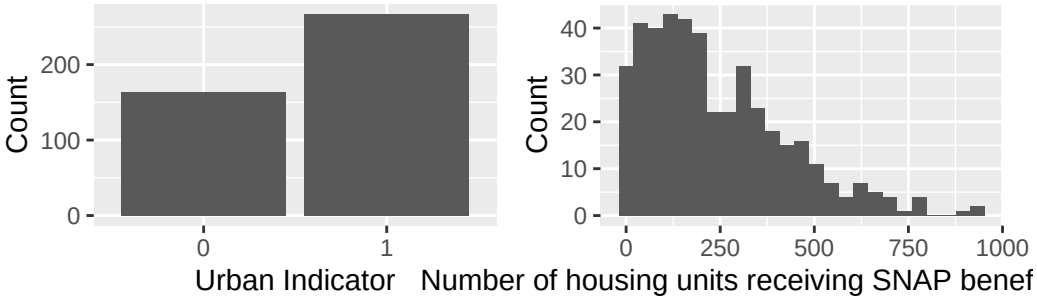
$$\text{LILATracts_1And10}_i = \beta_0 + \beta_1\,\text{Urban}_i + \beta_2\,\text{TractSNAP}_i + \beta_3\,\text{TractHUNV}_i + \beta_4\,\text{PovertyRate}_i + \beta_5\,\text{MedianFamilyI}$$

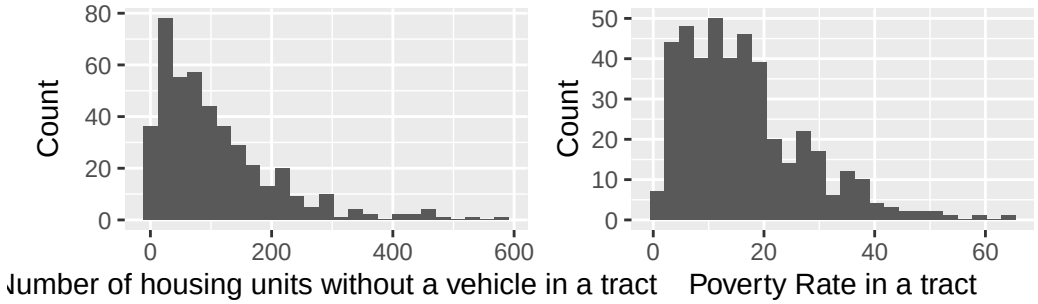
- Specific to NC
- take sample (.2*2175 = 435) to address independence
- took out missing values with `colSums(is.na(food_access_clean))` - only had to delete 4 obs (median fam income)




```
# A tibble: 2 x 3
  LILATracts_1And10     n  prop
  <fct>             <int> <dbl>
1 0                   363 0.842
2 1                    68 0.158
```








```
 PovertyRate      TractSNAP       TractHUNV     MedianFamilyIncome
Min.   : 0.70   Min.   :  0.0   Min.   :  0.0   Min.   : 17404
1st Qu.: 8.25   1st Qu.: 92.5   1st Qu.: 34.0   1st Qu.: 48536
```

```
Median :14.30   Median :188.0   Median : 80.0   Median : 61579
Mean   :16.35   Mean   :238.3   Mean   :106.2   Mean   : 68547
3rd Qu.:21.25   3rd Qu.:347.5   3rd Qu.:145.0   3rd Qu.: 81984
Max.   :64.20   Max.   :934.0   Max.   :580.0   Max.   :186310
```

term	estimate	std.error	statistic	p.value
(Intercept)	0.974	1.429	0.681	0.496
Urban1	2.685	0.468	5.739	0.000
TractSNAP	0.004	0.001	3.082	0.002
TractHUNV	-0.004	0.002	-1.894	0.058
MedianFamilyIncome	0.000	0.000	-4.412	0.000
PovertyRate	-0.038	0.024	-1.555	0.120

This section includes the following:

- Description of the data set and key variables.
- Exploratory data analysis of the response variable and key predictor variables. This includes visualizations, summary statistics, and narrative. Focus on 4 - 6 key predictors that are most relevant to your research question. You may explore additional variables, as appropriate, but prioritize depth over breadth.
  - Univariate EDA of the response and key predictor variables
  - Bivariate EDA of the response and key predictor variables
  - Visualizations exploring at least one potential interaction effect

For each visualization and/or set of summary statistics, briefly describe what you learn about the distribution of the variable or relationship between variables. Additionally, comment on how the observations from the EDA may inform your modeling decisions (e.g., use of transformations, quadratic effects, interactions effects, etc.)